

HOUSE CLUSTERING USING K-MEANS CLUSTERING TECHNIQUES

Project Report

Submitted by

Nikhil B Nair & Khushi Pragya

Abstract

House clustering is an unsupervised machine learning technique used to group houses with similar characteristics into meaningful clusters.

In this project, the K-Means Clustering algorithm was implemented to segment houses based on multiple attributes including area, number of bedrooms, bathrooms, floors, year built, location, condition, garage availability, and price. Before clustering, categorical variables were encoded using Label Encoding, and all numerical features were standardized using StandardScaler to ensure equal contribution during distance calculations.

The optimal number of clusters was determined using the Elbow Method, which indicated four suitable clusters for the housing dataset. The trained K-Means model successfully grouped houses into four distinct clusters representing different categories of residential properties. Cluster centers were analyzed to understand the average characteristics of each group, while Principal Component Analysis (PCA) was used to visualize the clustering results in two dimensions.

Introduction

The real estate industry generates large volumes of housing data containing information such as property size, number of bedrooms, location, construction year, condition, and market price. Analyzing these data manually becomes difficult as the number of properties increases. Machine learning techniques provide an efficient solution by automatically discovering patterns and grouping similar houses.

Clustering is one of the most widely used unsupervised machine learning techniques for exploratory data analysis. It divides data into groups such that objects within the same cluster are more similar to one another than to those in different clusters. Among various clustering algorithms, K-Means Clustering is popular because of its simplicity, computational efficiency, and effectiveness on structured datasets.

In this project, K-Means Clustering was applied to a housing dataset containing multiple numerical and categorical attributes. The objective was to group similar houses into

four meaningful clusters after appropriate preprocessing and feature scaling. The clustering results were further analyzed using cluster summaries, box plots, and PCA-based visualization to understand the characteristics of each group and evaluate the effectiveness of the algorithm.

Objective

To group similar houses into meaningful clusters using the K-Means clustering algorithm based on their physical and market-related features.

Dataset Description

The dataset used in this project consists of **2,000 residential house records** with **10 attributes**, including both numerical and categorical features. These attributes describe the physical characteristics and market-related information of each house. The dataset was selected to demonstrate the application of unsupervised machine learning techniques for grouping similar houses.

Dataset Features

Feature	Description
Id	Unique identifier for each house
Area	Total area of the house (square feet)
Bedrooms	Number of bedrooms
Bathrooms	Number of bathrooms
Floors	Number of floors
YearBuilt	Construction year of the house
Location	Location category of the house
Condition	Overall condition of the house
Garage	Availability of garage
Price	Selling price of the house

Methodology

The project followed a systematic machine learning workflow consisting of data preprocessing, feature engineering, clustering, visualization, and analysis.

The housing dataset was imported into the Python environment using the Pandas library.

The dataset was explored to understand:

- Number of rows and columns
- Data types
- Missing values
- Feature distribution

The dataset contained **2,000 observations** and **10 attributes**, with no missing values.

Data Preprocessing

The **Id** column was removed because it serves only as an identifier and has no influence on clustering.

Label Encoding

The following categorical variables were converted into numerical values using **LabelEncoder**:

- Location
- Condition
- Garage

Encoding allowed the clustering algorithm to process categorical information efficiently.

Feature Scaling

Since K-Means relies on Euclidean distance, feature scaling is essential to prevent variables with larger numerical ranges from dominating the clustering process.

The **StandardScaler** was used to standardize every feature.

This ensured equal contribution from every feature during distance calculations.

K-Means Clustering

K-Means is one of the most widely used unsupervised machine learning algorithms for grouping similar observations. The algorithm partitions the dataset into **K clusters**, where each observation belongs to the nearest centroid.

The working process of K-Means consists of the following steps:

1. Select the desired number of clusters (K).
2. Randomly initialize K centroids.
3. Assign each house to its nearest centroid.
4. Update each centroid by calculating the mean of all assigned houses.
5. Repeat the assignment and update steps until the centroids no longer change significantly.

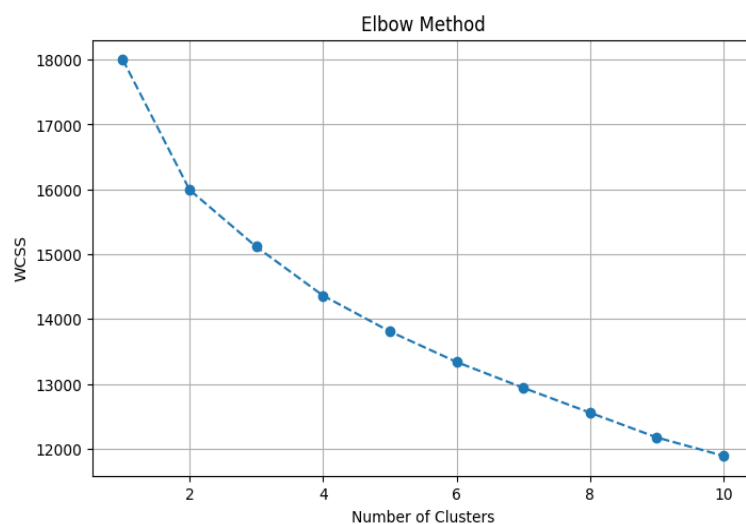
The algorithm minimizes the **Within Cluster Sum of Squares (WCSS)**, resulting in compact and meaningful clusters.

Determination of Optimal Number of Clusters

The **Elbow Method** was used to determine the optimal value of K.

The WCSS value was calculated for cluster sizes ranging from **1 to 10**. The resulting graph showed a significant decrease in WCSS until **K = 4**, after which the improvement became gradual.

Therefore, **four clusters** were selected for this project as they provided a good balance between cluster compactness and model complexity.

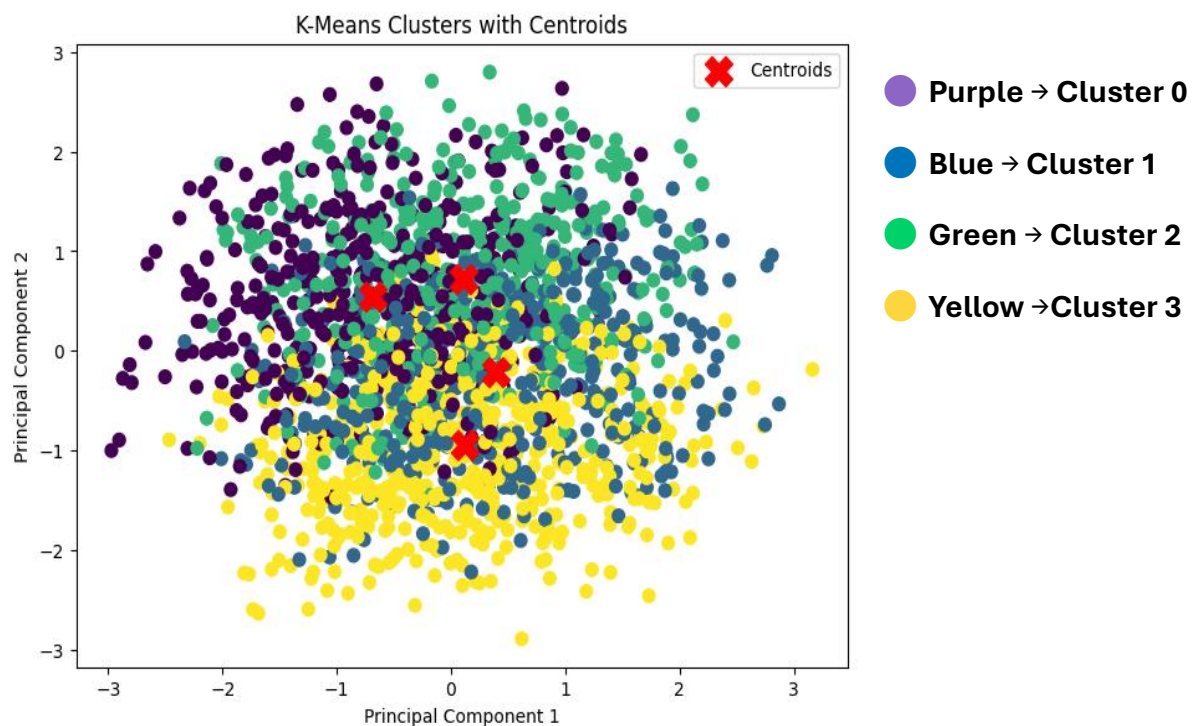


Principal Component Analysis (PCA)

Since the housing dataset contains multiple dimensions, direct visualization of clusters is difficult.

Principal Component Analysis (PCA) was applied to reduce the dataset to two principal components while preserving most of the important information. These components were used only for visualization purposes and did not affect the clustering process.

The PCA plot clearly illustrates the distribution of the four clusters along with their centroids.



Number of houses in each cluster

Cluster	Houses
3	543
2	504
1	495
0	458

The dataset was segmented into four clusters, with each cluster representing houses having similar physical and market-related characteristics. The distribution of houses among the clusters indicates that the algorithm successfully partitioned the dataset into meaningful groups.

The cluster centers represent the average values of each feature within a cluster and provide an overview of the typical characteristics of houses belonging to that group.

The analysis of **cluster centers** revealed the following:

- **Cluster 0** consisted of houses with the lowest average price (approximately **₹312,543**), representing affordable or budget residential properties.
- **Cluster 1** contained houses with an average price of approximately **₹541,019**, representing mid-range residential properties with moderate house characteristics.
- **Cluster 2** exhibited the highest average selling price (approximately **₹743,820**), indicating premium or luxury residential properties with superior overall characteristics.
- **Cluster 3** showed an average price of approximately **₹533,185**. Although its average price was close to Cluster 1, differences in other attributes such as location, number of floors, garage availability, condition, and year built resulted in a separate cluster.

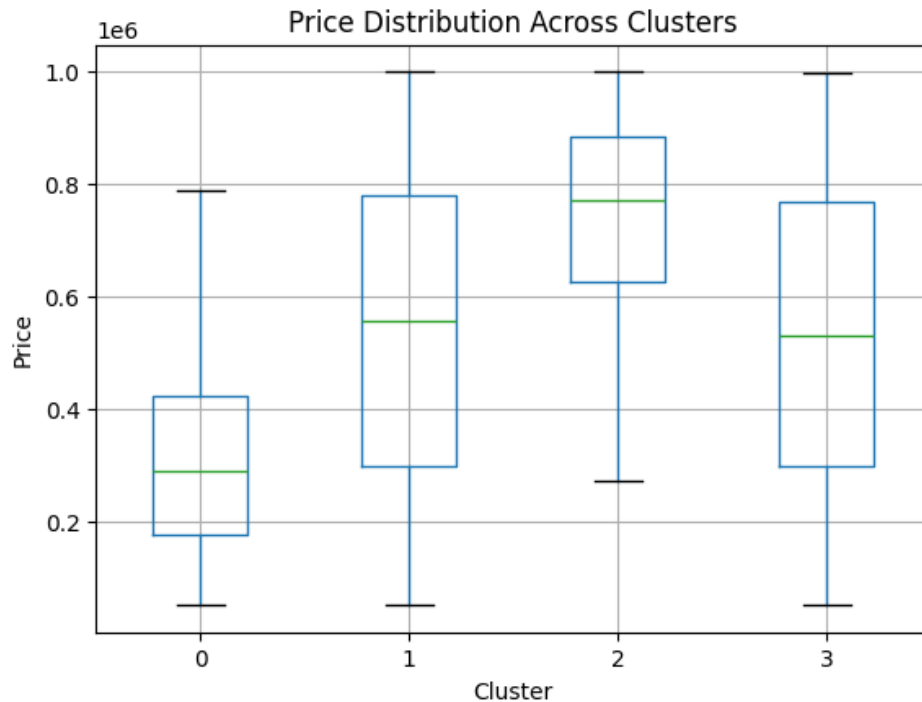
Price Distribution Across Clusters

A box plot was generated to compare the distribution of house prices across the four clusters.

The box plot indicates noticeable variation in house prices among the clusters. Cluster 2 contains the highest-priced houses, whereas Cluster 0 primarily consists of lower-priced properties. Clusters 1 and 3 represent intermediate price ranges with moderate variation.

This visualization further validates that the clustering process successfully grouped houses into different market segments.

The boxplot is given below.



Key Insights

The experimental analysis led to the following observations:

- The housing dataset contained **2,000 records** and **10 attributes** with no missing values.
- Three categorical features were successfully converted into numerical form using Label Encoding.
- Feature scaling ensured equal importance of all variables during distance computation.
- The Elbow Method identified **four** as the optimal number of clusters.
- K-Means successfully grouped similar houses into four meaningful categories.
- PCA visualization confirmed the presence of distinguishable clusters.
- Cluster 2 represented premium houses with the highest average selling price.
- Cluster 0 represented affordable houses with the lowest average price.
- Clusters 1 and 3 contained houses with similar price ranges but differed in other structural and location-based characteristics.

Conclusion

This project successfully demonstrated the application of the **K-Means Clustering algorithm** for grouping similar houses based on multiple housing characteristics.

The complete machine learning workflow included data preprocessing, label encoding, feature scaling, determination of the optimal number of clusters using the Elbow Method, clustering, visualization through Principal Component Analysis, and interpretation of cluster characteristics.

The algorithm effectively segmented the housing dataset into four meaningful clusters representing different categories of residential properties. The analysis showed that clustering based on multiple features provides deeper insights than considering a single attribute such as price alone.

Overall, the project highlights the usefulness of unsupervised machine learning techniques in exploratory data analysis, customer segmentation, and real estate market analysis.

REFERENCES

House price prediction Dataset from Kaggle

<https://www.kaggle.com/datasets/zafarali27/house-price-prediction-dataset>